

Executive Summary

ATM Cash Utilization & Downtime Analysis

Project Overview

This project evaluates ATM transaction volumes, cash demand, utilization patterns and potential cash availability risk using Python-based data analysis and SQL-based reporting.

Key Metrics

Metric	Value
ATM Records	5,658
Distinct ATMs	50
Total Withdrawals	281,814,269
Total Deposits	57,310,326
Average Utilization	52.05%
High-Risk Records	363

Major Findings

- The Bank Branch location category records the highest aggregate withdrawals with approximately 57,507,375 withdrawal transactions.
- The highest average next-day cash demand is observed in the Bank Branch location category, at approximately 43,617.28.
- The highest average cash utilization is observed in the Supermarket location category at 53.12%.
- Afternoon records the highest aggregate withdrawal activity among the available time-of-day categories.
- Thursday records the highest aggregate withdrawal activity among the available days of the week.

Conclusion

The analysis demonstrates how ATM transaction data can be transformed into actionable metrics for cash utilization monitoring, demand analysis and potential cash availability risk assessment.

Note: Cash_Risk represents an analytical risk indicator and should not be interpreted as confirmed ATM downtime.